

Fractions (Bhinn)

Hinglish Notes, Types, Operations aur Step-by-Step Solutions

Fraction kisi poori cheez ke hisse (part) ko represent karta hai. Is guide me fraction kya hai, uske types, simplest form, comparison aur sabhi operations (jod-ghata-guna-bhaag) simple Hinglish me samjhaye gaye hain, aur end me 22 practice questions ke STEP-BY-STEP solutions diye gaye hain.

1. Fraction kya hota hai?

Definition Fraction kisi cheez ke barabar hisson me se kuch hisson ko batata hai. Ise p/q form me likha jata hai.

Example: $3/4$ ka matlab - poori cheez ko 4 barabar hisson me baanta, aur usme se 3 hisse liye.

Parts:

- Numerator (ansh) = upar wala number (kitne hisse liye). Jaise $3/4$ me 3.
- Denominator (har) = neech wala number (total kitne hisse). Jaise $3/4$ me 4.
- Denominator kabhi 0 nahi ho sakta.

2. Fractions ke Types

- Proper Fraction: numerator < denominator. Jaise $3/4$, $2/5$.
- Improper Fraction: numerator $\geq$ denominator. Jaise $7/4$, $5/5$.
- Mixed Fraction: whole number + proper fraction. Jaise $1\frac{3}{4}$.
- Like Fractions: same denominator. Jaise $2/5$ aur $3/5$.
- Unlike Fractions: alag denominator. Jaise $1/2$ aur $1/3$.
- Equivalent Fractions: value barabar. Jaise $1/2 = 2/4 = 3/6$.
- Unit Fraction: numerator 1 ho. Jaise $1/5$, $1/8$.

3. Equivalent Fractions aur Simplest Form

Equivalent banane Numerator aur denominator dono ko same number se multiply ya divide karo. Jaise $1/2 = (1 \times 2)/(2 \times 2) = 2/4$.

Simplest Form Numerator aur denominator ko unke HCF se divide karo jab tak aur simplify na ho sake.

- Example: $6/9 \rightarrow$ HCF 3 $\rightarrow (6/3)/(9/3) = 2/3$ (simplest form).

4. Fractions ki Comparison

- Same denominator ho to bada numerator wala fraction bada hota hai. Jaise $3/5 > 2/5$.
- Alag denominator ho to LCM lekar same denominator banao, fir compare karo.
- Same numerator ho to chhota denominator wala bada hota hai. Jaise $1/2 > 1/3$.

5. Fractions par Operations

(a) Addition / Subtraction Same denominator ho to numerators jodo/ghatao, denominator wahi rakho. Alag ho to pehle LCM se same banao.

- Example: $1/2 + 1/3 \rightarrow$ LCM 6 $\rightarrow 3/6 + 2/6 = 5/6$

(b) Multiplication Numerator x numerator, denominator x denominator. Fir simplify karo.

- Example: $2/3 \times 3/4 = 6/12 = 1/2$

(c) Division Doosre fraction ko ulta (reciprocal) karke multiply karo.

- Example: $1/2 \div 1/4 = 1/2 \times 4/1 = 4/2 = 2$

6. Conversions

Improper -> Mixed Numerator ko denominator se divide karo. Quotient = whole part, remainder/denominator = fraction part.

- Example: $7/2 \rightarrow 7 \div 2 = 3$ remainder 1 $\rightarrow 3 \frac{1}{2}$

Mixed -> Improper (whole $\times$ denominator) + numerator, denominator wahi.

- Example: $2 \frac{1}{3} \rightarrow (2 \times 3) + 1 = 7 \rightarrow 7/3$

Fraction -> Decimal Numerator ko denominator se divide karo. Jaise $1/4 = 0.25$.

Quick Revision

Proper: upar < neeche ($3/4$) | Improper: upar $\geq$ neeche ($7/4$)

Add/Subtract: denominator same karo (LCM).

Multiply: seedha upar-upar, neeche-neeche.

Divide: doosre ko ulta karke multiply.

Simplest form: HCF se divide.

Solved Examples - Samjho Kaise Solve Karte Hain

Neeche 10 examples step-by-step solve karke dikhaye gaye hain, taaki har type ka method clear ho jaye (jod, ghata, guna, bhaag, negative, mixed).

Example 1: $2/3 + 1/4$ (unlike addition)

- Step: Denominator alag hain $\rightarrow$ LCM(3, 4) = 12.
- Step: Dono ko 12 wale denominator me badlo:
- Step: $2/3 = 8/12$ (upar-neeche $\times 4$), $1/4 = 3/12$ ($\times 3$).
- Step: Numerators jodo: $8/12 + 3/12 = 11/12$.

Answer: $11/12$

Example 2: $5/6 - 1/4$ (unlike subtraction)

- Step: LCM(6, 4) = 12.
- Step: $5/6 = 10/12$ ($\times 2$), $1/4 = 3/12$ ($\times 3$).
- Step: Ghatao: $10/12 - 3/12 = 7/12$.

Answer: $7/12$

Example 3: $2/3 \times 4/5$ (multiplication)

- Step: Upar $\times$ upar: $2 \times 4 = 8$.
- Step: Neeche $\times$ neeche: $3 \times 5 = 15$.
- Step: $= 8/15$. HCF(8,15)=1, isliye already simplest.

Answer: $8/15$

Example 4: $2/4 \times 6/5 \times -2/7$ (negative, 3 fractions)

- Step: Sign dekho: ek hi negative hai $\rightarrow$ answer NEGATIVE hoga.
- Step: Numerators multiply: $2 \times 6 \times 2 = 24$.
- Step: Denominators multiply: $4 \times 5 \times 7 = 140$.
- Step: $= -24/140$.

- Step: Simplify: $\text{HCF}(24,140) = 4 \rightarrow -24/4 / 140/4$.

Answer: $-6/35$

Example 5: $3/4 \div 2/5$ (division)

- Step: Division me doosre ka reciprocal lo: $2/5 \rightarrow 5/2$.
- Step: Ab multiply: $3/4 \times 5/2 = 15/8$.
- Step: $15/8$ improper hai $\rightarrow$ mixed = $1 \frac{7}{8}$.

Answer: $15/8$ (ya $1 \frac{7}{8}$)

Example 6: $1 \frac{1}{2} + 2 \frac{1}{3}$ (mixed addition)

- Step: Pehle improper banao: $1 \frac{1}{2} = 3/2$, $2 \frac{1}{3} = 7/3$.
- Step: $\text{LCM}(2,3) = 6 \rightarrow 3/2 = 9/6$, $7/3 = 14/6$.
- Step: Jodo: $9/6 + 14/6 = 23/6$.
- Step: Mixed me: $23/6 = 3 \frac{5}{6}$.

Answer: $23/6$ (ya $3 \frac{5}{6}$)

Example 7: $18/24$ ko simplest form me likho

- Step: $\text{HCF}(18, 24) = 6$ nikalo.
- Step: Upar-neeche ko 6 se divide: $18/6 = 3$, $24/6 = 4$.

Answer: $3/4$

Example 8: $3/4$ aur $5/6$ me bada kaunsa? (compare)

- Step: $\text{LCM}(4, 6) = 12$.
- Step: $3/4 = 9/12$, $5/6 = 10/12$.
- Step: $10/12 > 9/12$.

Answer: $5/6$ bada hai

Example 9: $-3/5 \times 10$ (negative x whole number)

- Step: 10 ko fraction maano: $10/1$.
- Step: $-3/5 \times 10/1 = -30/5$.
- Step: $-30/5 = -6$.

Answer: -6

Example 10: 20 ka $3/4$ (kisi quantity ka fraction)

- Step: 'ka' / 'of' ka matlab multiply hota hai.
- Step: $3/4 \times 20 = 3/4 \times 20/1 = 60/4$.
- Step: $60/4 = 15$.

Answer: 15

Practice Questions

Pehle khud solve karne ki koshish karo. Neeche har question ka step-by-step solution diya gaya hai.

Q1. $3/4$ me numerator kaunsa number hai?

Q2. $7/4$ kis type ka fraction hai (proper / improper)?

Q3. Kya $2/3$ aur $4/6$ equivalent fractions hain?

Q4. $5/10$ ko simplest form me likho.

Q5. $1/2 + 1/2 = ?$

Q6. $3/5 + 1/5 = ?$

Q7. $1/2 + 1/3 = ?$

- Q8. $\frac{3}{4} - \frac{1}{4} = ?$
- Q9. $\frac{2}{3} \times \frac{3}{4} = ?$
- Q10. $\frac{1}{2} \div \frac{1}{4} = ?$
- Q11. $\frac{7}{2}$ ko mixed fraction me likho.
- Q12. $2 \frac{1}{3}$ ko improper fraction me likho.
- Q13. $\frac{1}{4}$ ko decimal me likho.
- Q14. Proper fraction ki definition kya hai?
- Q15. Unit fraction kya hota hai? Ek example do.
- Q16. Kya $\frac{4}{8}$, $\frac{1}{2}$ aur $\frac{3}{6}$ sab equivalent hain?
- Q17. $\frac{2}{5}$ aur $\frac{3}{5}$ me se bada kaunsa hai?
- Q18. $\frac{1}{2}$ aur $\frac{1}{3}$ me se bada kaunsa hai?
- Q19. $\frac{6}{9}$ ka simplest form kya hai?
- Q20. $\frac{3}{4} \times 4 = ?$
- Q21. Mixed fraction kya hota hai? Ek example do.
- Q22. 0.5 ko fraction me likho.

Step-by-Step Solutions

Q1. $\frac{3}{4}$ me numerator?

- Step: Fraction me upar wala number numerator hota hai.
- Step: $\frac{3}{4}$ me upar 3 hai.

Answer: Numerator = 3

Q2. $\frac{7}{4}$ ka type?

- Step: Numerator = 7, Denominator = 4.
- Step: Yahan numerator (7) > denominator (4).

Answer: Improper Fraction

Q3. $\frac{2}{3}$ aur $\frac{4}{6}$ equivalent?

- Step: $\frac{2}{3}$ ke upar-neeche ko 2 se multiply karo.
- Step: $(2 \times 2) / (3 \times 2) = \frac{4}{6}$.
- Step: Dono ki value barabar nikli.

Answer: Haan, equivalent hain

Q4. $\frac{5}{10}$ simplest form?

- Step: HCF(5, 10) = 5 nikalo.
- Step: Upar-neeche ko 5 se divide: $(5/5) / (10/5)$.

Answer: $\frac{1}{2}$

Q5. $\frac{1}{2} + \frac{1}{2} = ?$

- Step: Denominator same (2) hai, to numerators jodo.
- Step: $(1+1)/2 = \frac{2}{2}$.
- Step: $\frac{2}{2}$ simplify $\rightarrow 1$.

Answer: 1

Q6. $\frac{3}{5} + \frac{1}{5} = ?$

- Step: Denominator same (5) hai.
- Step: Numerators jodo: $(3+1)/5$.

Answer: $\frac{4}{5}$

Q7. $1/2 + 1/3 = ?$

- Step: Denominator alag hain $\rightarrow$ LCM(2,3) = 6.
- Step: $1/2 = 3/6$ aur $1/3 = 2/6$.
- Step: Ab jodo: $3/6 + 2/6$.

Answer: $5/6$

Q8. $3/4 - 1/4 = ?$

- Step: Denominator same (4) hai.
- Step: Numerators ghatao: $(3-1)/4 = 2/4$.
- Step: Simplify: HCF 2 se $\rightarrow 1/2$.

Answer: $2/4 = 1/2$

Q9. $2/3 \times 3/4 = ?$

- Step: Multiplication: upar x upar, neeche x neeche.
- Step: $(2 \times 3)/(3 \times 4) = 6/12$.
- Step: Simplify: HCF 6 se $\rightarrow 1/2$.

Answer: $6/12 = 1/2$

Q10. $1/2 \div 1/4 = ?$

- Step: Division me doosre ka reciprocal lo: $1/4 \rightarrow 4/1$.
- Step: Ab multiply: $1/2 \times 4/1 = 4/2$.
- Step: $4/2$ simplify $\rightarrow 2$.

Answer: 2

Q11. $7/2 \rightarrow$ mixed fraction?

- Step: Numerator ko denominator se divide: $7 \div 2$.
- Step: Quotient = 3, Remainder = 1.
- Step: Mixed = quotient + (remainder/denominator).

Answer: $3 \frac{1}{2}$

Q12. $2 \frac{1}{3} \rightarrow$ improper fraction?

- Step: Formula: (whole x denominator) + numerator.
- Step: $(2 \times 3) + 1 = 7$.
- Step: Denominator wahi (3) rakho.

Answer: $7/3$

Q13. $1/4 \rightarrow$ decimal?

- Step: Numerator ko denominator se divide: $1 \div 4$.

Answer: 0.25

Q14. Proper fraction definition?

- Step: Jis fraction me numerator chhota ho denominator se.

Answer: Numerator < Denominator (jaise $3/4$)

Q15. Unit fraction kya hai?

- Step: Wo fraction jiska numerator 1 ho.

Answer: Jaise $1/5$, $1/7$

Q16. $4/8$, $1/2$, $3/6$ equivalent?

- Step: $4/8$ ko HCF 4 se simplify $\rightarrow 1/2$.
- Step: $3/6$ ko HCF 3 se simplify $\rightarrow 1/2$.
- Step: Teeno ki value $1/2$ nikli.

Answer: Haan, sab equivalent

Q17. $2/5$ aur $3/5$ me bada?

- Step: Denominator same (5) hai.
- Step: Bada numerator wala bada: $3 > 2$.

Answer: $3/5$ bada hai

Q18. $1/2$ aur $1/3$ me bada?

- Step: Numerator same (1) hai.
- Step: Chhota denominator wala bada: $2 < 3$.

Answer: $1/2$ bada hai

Q19. $6/9$ simplest form?

- Step: $HCF(6, 9) = 3$ nikalo.
- Step: Upar-neeche ko 3 se divide: $(6/3)/(9/3)$.

Answer: $2/3$

Q20. $3/4 \times 4 = ?$

- Step: 4 ko fraction maano: $4/1$.
- Step: $3/4 \times 4/1 = 12/4$.
- Step: Simplify: $12/4 = 3$.

Answer: 3

Q21. Mixed fraction kya hai?

- Step: Whole number + proper fraction milkar banta hai.

Answer: Jaise $2 \frac{1}{3}$

Q22. 0.5 -> fraction?

- Step: $0.5 = 5/10$ likho.
- Step: $HCF 5$ se simplify: $(5/5)/(10/5)$.

Answer: $1/2$